



Insomnia

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Sleep is a paradox



Sleep Disorders

- Insomnias
- Sleep Apnea
- Parasomnias
- Circadian Disorders
- Restless Leg Syndrome
- Hypersomnia/ Narcolepsy



- Why do people get insomnia?
- How does it become chronic?
- How do we fix it?

Insomnia

Trouble falling asleep or staying asleep that results in some form of daytime impairment. It is a symptom that can become a syndrome.

ICSD-3 subtypes:

- Chronic Insomnia Disorder:
symptoms occur at least three times per week and present for at least three months
- Short Term Insomnia Disorder: also referred to as acute or adjustment insomnia
- Other Insomnia Disorder: provisional diagnosis



- Why do people get insomnia?
- How does it become chronic?
- How do we fix it?

Nighthawks
Edward Hopper
1942



The need for sleep is biological,

The way you sleep is learned

What wakes you up, *may not be what keeps you awake*

If you tell your doctor you are tired and
your doctor say they are tired too, you
are seeing the wrong doctor



Most p
b



will get
ly

Insomnia

- On any given night half of us don't think we slept well
- One adult out of 4 complains of insomnia; however only 1 insomniac out of 4 has ever complained about it to their doctor during a visit made for another problem and only 1 out of 20 has come to discuss specifically insomnia.
- About 12-15% of people have chronic insomnia

What's like to live with chronic insomnia?

- Fatigue/ malaise
- Attention, concentration, memory impairment
- Impaired social, academic, occupational performance
- Mood Disturbances
- Daytime sleepiness
- Behavioral problems
- Decreased motivation

**Insomniacs don't sleep
with other insomniacs!**



Insomnia

- Insomnia has been found to be associated with decreased quality of life and productivity
- Increased absenteeism from work; insomniacs miss work twice as often as non-insomniacs
- Increased work accidents
- Increased general healthcare utilization
- It has also been identified as a risk factor for the subsequent development of other conditions such as major depression, generalized anxiety disorder, and coronary artery disease



Insomnia Economics



Estimates of total insomnia-related costs in the United States have ranged from \$30 to \$35 billion per year to \$92.5 to \$107.5 billion per year. These costs include direct treatment costs, such as physician encounters and prescriptions, as well as indirect costs, such as consumption of medical services, increased accident risk, and lost workplace productivity.

Source: *Sleep Review* 12/14

Insomnia Impact on Society

- Employees with insomnia report more social isolation, and decreased energy, emotional reaction, and physical mobility
- Insomniac patients rate more poorly on the SF-36 for QOL in areas of social functioning, physical functioning, vitality, and mental health
- Patients with insomnia were found to have more deficits than those with major depression or congestive heart failure on scales including general health and vitality

What Drives Us
Toward Sleep and
What Keeps Us
Asleep?



People with insomnia
talk a certain way.
They talk about
“trying” to sleep and
being unable to “turn
off” their brains

Sleep Timing

- Sleep timing is influenced by homeostatic and circadian factors
- The less we sleep the more sleep we need and vice versa
- Twice a day our alertness level peaks
- Twice a day our sleepiness peaks

Sleep Hypervigilance Model

Predisposing Factors

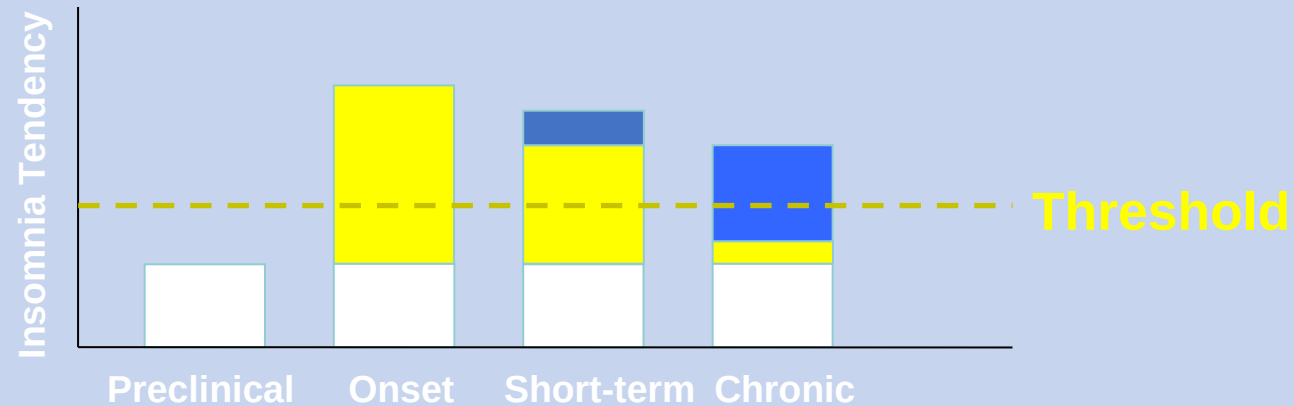
- Biological traits
- Psychological traits
- Social factors

Precipitating Factors

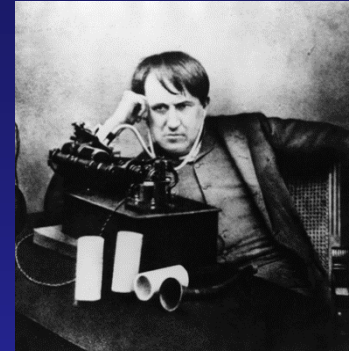
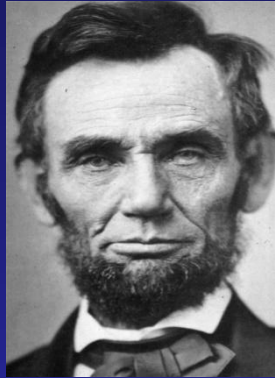
- Medical illness
- Psychiatric illness
- Stressful life events

Perpetuating Factors

- Excessive time in bed
- Napping
- Conditioning



This durable model was developed here in NY at City College by the late Arthur Spielman



The thought of sleeping wakes them up



Sleep hygiene rules often recommend for healthy sleep

- Avoid napping during the day
- Avoid stimulants such as caffeine, nicotine, and alcohol too close to bedtime
- Exercise can promote good sleep
- Food can be disruptive right before sleep
- Establish a regular relaxing bedtime routine
- Associate your bed with sleep

Sole reliance on sleep hygiene is usually ineffective in the treatment of chronic insomnia. Insomniacs may have these recommendations memorized and say they do not work for them. This may be due to their conditioned hypervigilance of sleeping



Who Likes Relying On **Sleeping Pills?**



If there is a stigma, why are they popular?

The Real Villain Is Not The Pill, But The **Insomnia**



Patients may complain to you about their sleep but what they are really complaining about is how they feel when they are awake which they are attributing to their sleep!



Paradoxical Reaction To A Hypnotic Medication



Over-the-Counter Sleep Aids

- Regulated by FDA
- Antihistamines (branded or generic)
 - Diphenhydramine
 - Doxylamine
- Alone or combined with analgesics
- Pharmacodynamic action: postsynaptic histaminic and muscarinic blockade
- Sedating effect

Melatonin Natural Health Products and Supplements: Presence of Serotonin and Significant Variability of Melatonin Content *JCSM'17*

- 31 supplements were analyzed by ultraperformance liquid chromatography
- “Melatonin content was found to range from –83% to +478% of the labelled content. Additionally, lot-to-lot variable within a particular product varied by as much as 465%. This variability did not appear to be correlated with manufacturer or product type. Furthermore, serotonin (5-hydroxytryptamine), a related indoleamine and controlled substance used in the treatment of several neurological disorders, was identified in eight of the supplements at levels of 1 to 75 µg”
- Melatonin content did not meet label within a 10% margin of the label claim in more than 71% of supplements and an additional 26% were found to contain serotonin



**Do Not To Equate
Sedation With
Normal Sleep!**



Clinical Practice Guideline for the Pharmacologic Treatment of Chronic Insomnia in Adults: An American Academy of Sleep Medicine Clinical Practice Guideline JCSM '17

Recommendation “weak” for:

Use: Suvorexant, eszopiclone, zaleplon, zolpidem, triazolam, temazepam, ramelteon, doxepin

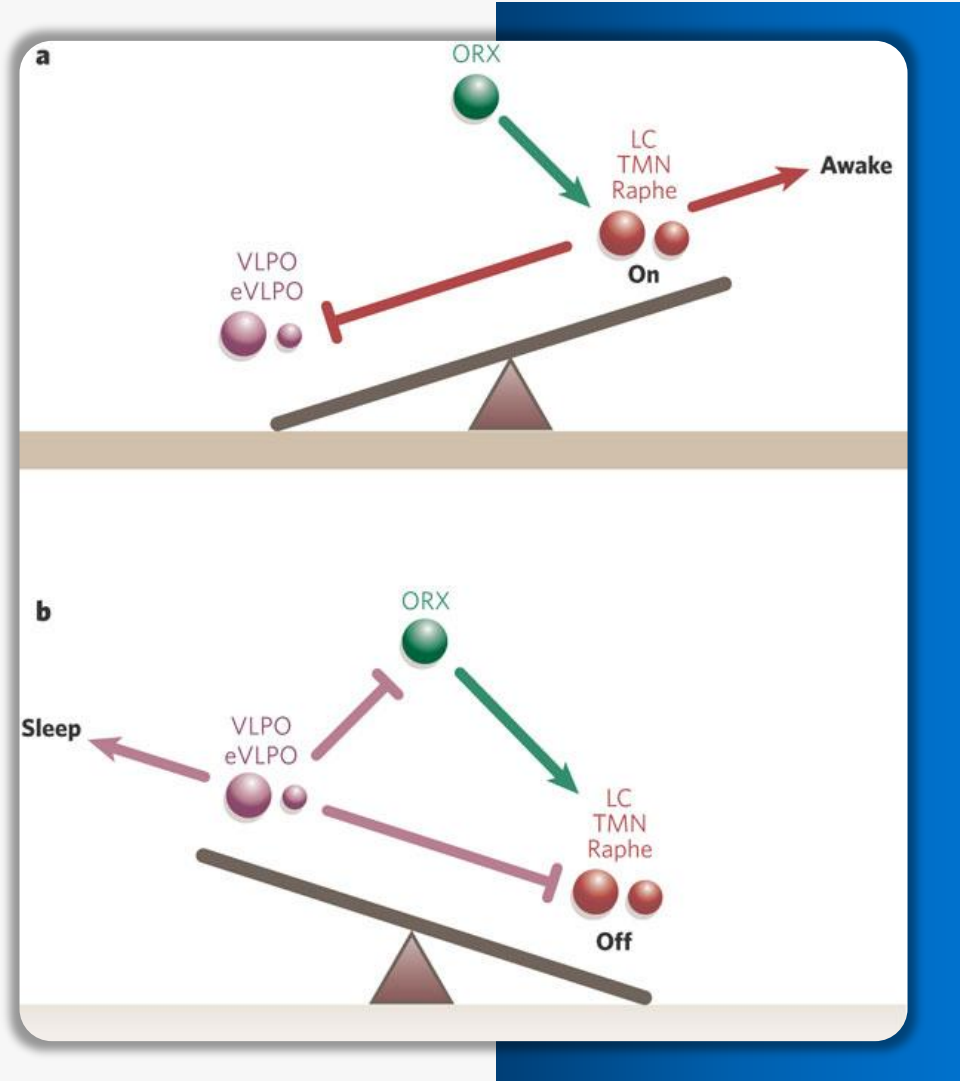
Not Use: trazodone, tiagabine, diphenhydramine, melatonin, tryptophan, valerian

“A WEAK recommendation reflects a lower degree of certainty in the outcome and appropriateness of the patient-care strategy for all patients, but should not be construed as an indication of ineffectiveness”

Other agents Medical Letter 2018

HERBAL PRODUCTS — “**Valerian root** is claimed to be a mild hypnotic that may improve the quality of sleep, but some studies have found it ineffective. There is no convincing evidence that any of the other “natural” remedies used for insomnia, such as **kava** or **lavender**, are effective or safe for this indication and the purity and optimal dosages of all these products are unknown. No objective data have been published to date to support the subjective claim of sleep improvement attributed to **cannabis** or its various extracts.”

Orexin/ Hypocretin Sleep Flip Flop Switch



Dual Orexin Receptor Antagonists

Orexin/ hypocretin promotes and stabilizes wakefulness with projections from the lateral hypothalamus to the cortex and to the nuclei of multiple wake-promoting neurotransmitters (histamine, acetylcholine, dopamine, serotonin, and norepinephrine).

- These medications have antagonist activity at both of the orexin (hypocretin) receptors, thereby facilitating sleep by decreasing the wake drive.
- Narcolepsy like side effects:
 - ✓ Cataplexy
 - ✓ EDS
 - ✓ Impaired driving
 - ✓ Unconscious nighttime behaviors
 - ✓ Suicidal ideation



Manufacturer Files Petition with DEA Urging Them to De-Schedule Dual Orexin Receptor Antagonists Medications **April 4, 2023**

A Citizen Petition requests that the DEA initiate a rulemaking proceeding to remove the DORA class medications from scheduling under the Controlled Substances Act based on the following:



8 yrs of post-marketing surveillance data suggests that they have “an insignificant risk of abuse profile and potential for abuse, lack of non-medical use in the community; and similar lack of physical or psychological dependence”



“Based on major federal surveillance systems including FDA’s Adverse Events Reporting System the rates of abuse and dependence-related reports, serious adverse events (AEs), and AEs requiring hospitalization associated with the DORA class are extremely low in incidence and substantially lower than rates for Schedule IV benzodiazepines and related drugs...”



“A re-analysis of the Eight Factors of the Controlled Substances Act (a required step in the scheduling process) supports removal of the DORA class from scheduling under the CSA”

Management of Chronic Insomnia Disorder in Adults: A Clinical Practice Guideline From the American College of Physicians 2016

- The American College of Physicians “... *recommends that all adult patients receive cognitive behavioral therapy for insomnia (CBT-I) as the initial treatment for chronic insomnia disorder.*”
- Online versions of CBT for insomnia are available

**What is your motivation to
go to bed?**

**What is your motivation
to get out of bed?**

SELF Correction

- Social
- Exercise
- Light
- Food

Cognitive Behavioral Therapy

- Stimulus Control
 - Targets negative association between bed/bedroom and poor sleep
 - Re-associates bed/bedroom with rapid sleep onset
 - Involves procedures associated with bed/bedroom behavior
- Sleep Restriction
 - Restricts time in bed to match subjective total sleep time (TST)
 - Modifies allowed time in bed each week until target TST is reached
- Paradoxical Intention
 - Aims to eliminate performance anxiety associated with insomnia
 - Persuades patient to confront fear of staying awake
- Scheduled Thinking Time
 - Also called scheduled worry time
 - Works to help to control racing thoughts

Insomnia Summary

- Insomnia is a learned behavior that can be improved
- Insomniacs should not be stigmatized
- Hypnotics have gotten safer, not stronger
- Cognitive Behavioral Therapy should be the mainstay treatment
- Do not confuse marketing with reality

Questions?

